

16.1

$$\begin{aligned}
 a) \hat{P}(AUTO=1) &= \Lambda(\hat{\gamma}_1 + \hat{\gamma}_2 DTIME) \\
 &= \Lambda(-0,2376 + 0,5311) \\
 &= \Lambda(0,2935) \\
 &= \frac{1}{1 + e^{-0,2935}} = 0,5729
 \end{aligned}$$

b)

$$\tilde{\beta}_1 + \tilde{\beta}_2 DTIME_{(sc)} = -0,0644 + 0,3 DTIME_{(0,3992)} \quad (0,1029)$$

$$\begin{aligned}
 P(AUTO=1) &= \Phi(-0,0644 + 0,3 DTIME) \\
 &= \Phi(-0,0644 + 0,3 \times 1) \\
 &= \Phi(0,2356) \\
 &= 0,5931 > 0,5729
 \end{aligned}$$

⇒ This result is higher than logit result

c)

$$\begin{aligned}
 &\Lambda(-0,2376 + 0,5311 \times 3) \\
 &= \Lambda(1,3557) \\
 &= \frac{e^{-1,3557}}{(1 + e^{-1,3557})^2} \quad \left(\Lambda(1) = \frac{e^{-1}}{(1 + e^{-1})^2}, -\infty < 1 < \infty \right) \\
 &= 0,1629
 \end{aligned}$$

The estimated marginal effect:

$$\begin{aligned}
 \frac{dP}{dDTIME} &= \Lambda(z) [1 - \Lambda(z)] \gamma_2 \quad \left(z = -0,2376 + 0,5311 \times 3 = 1,3557 \right. \\
 &\quad \left. \Lambda(z) = \frac{1}{1 + e^{-1,3557}} = 0,7951 \right) \\
 &= 0,7951 [1 - 0,7951] \times 0,5311 \\
 &= 0,0865
 \end{aligned}$$

Use linear probability model from Ex 16.2:

$$\widehat{AUTO} = 0,4848 + 0,0703 DTIME$$

The marginal effect is 0,0703 < 0,0865

The logit marginal effect is larger than the linear model, but both imply that increasing 10 min in bus travel time related to travel time will increase the prob. of choosing automobile transportation.

d)

$$DTIME = -5$$

$$\text{Logit model: } z = -0,2376 + 0,5311 \times (-5) = -2,8931$$

$$\Lambda(-2,8931) = 0,0525$$

Marginal effect:

$$ME = 0,0525 (1 - 0,0525) \times 0,5311 = 0,0264 \rightarrow \text{decrease } 0,0264$$

$$\text{Probit model: } z = -0,0644 + 0,3 \times (-5) = -1,5644$$

$$\Phi(-1,5644) = 0,1173$$

$$ME = 0,1173 \times (0,3) = 0,0352 \rightarrow \text{decrease } 0,0352$$

16. h.

a)

$$P(y=1) = \Lambda(x_1 + x_2 \alpha)$$

$$\hat{\gamma}_1 + \hat{\gamma}_2 x = -1,836 + 3,021x$$

$$\text{With } x = 1,5 : z = -1,836 + 3,021(1,5) = 2,6955$$

$$P(y=1) = \Lambda(2,6955) = 0,9368$$

b)

Since $0,9368 > 0,5$, we predict that $y = 1$, which is consistent to the actual result

c)

$$x_1 = -1, \quad x_2 = 2$$

$$\text{Ex 16.3} \quad x_1 = 1,5 ; y_1 = 1$$

$$x_2 = 0,6 ; y_2 = 1$$

$$x_3 = 0,7 ; y_3 = 0$$

$$z = -1 + 2 \times 1,5 = 2 \rightarrow p_1 = \Lambda(2) = 0,8808$$

$$z = -1 + 2 \times 0,6 = 0,2 \rightarrow p_2 = \Lambda(0,2) = 0,5498$$

$$z = -1 + 2 \times 0,7 = 0,4 \rightarrow p_3 = \Lambda(0,4) = 0,5987$$

$$y_3 = 0 = 1 - p_3$$

$$L = p_1 p_2 (1 - p_3) = 0,8808 \cdot 0,5498 \cdot (1 - 0,5987)$$

The maximized log-likelihood is given at: $\log L(\hat{\gamma}_1, \hat{\gamma}_2) = -1,612$

$$\Rightarrow L(\hat{\gamma}_1, \hat{\gamma}_2) = e^{-1,612} = 0,1995$$

$\Rightarrow L(\hat{\gamma}_1, \hat{\gamma}_2)$ is larger ($0,1995 > 0,1943$)

d)

True. For each obs, the logit model gives a probability:

$$\begin{cases} 0 < \Lambda(z) < 1 \\ 0 < 1 - \Lambda(z) < 1 \end{cases}$$

The likelihood function is the product of these probabilities:

$$L(y_1, y_2) = \prod_{i=1}^n P_i(y_i=1)^{y_i} (1 - P(y_i=1))^{1-y_i}$$

Since each term is a probability btw 0 and 1, the result is also from 0 to 1.

$$\Rightarrow 0 < L(y_1, y_2) < 1$$

e) True

By (d), we know that the likelihood function is btw 0 and 1, so after taking the logarithm its value is always be negative.

a.

```
> #a
> lpm <- lm(delinquent ~ lvr + ref + insur + rate +
+           amount + credit + term + arm,
+           data = lasvegas)
>
> # White heteroskedasticity-robust SE
> coeftest(lpm, vcov = vcovHC(lpm, type = "HC1"))
```

t test of coefficients:

```
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.68849128  0.22850643   3.0130 0.0026525 **
lvr           0.00162385  0.00067523   2.4049 0.0163601 *
ref          -0.05932367  0.02402561  -2.4692 0.0137098 *
insur        -0.48158489  0.03036944 -15.8575 < 2.2e-16 ***
rate          0.03437614  0.00981944   3.5008 0.0004845 ***
amount        0.02376801  0.01445089   1.6447 0.1003398
credit       -0.00044190  0.00020728  -2.1319 0.0332620 *
term         -0.01261951  0.00355596  -3.5488 0.0004051 ***
arm           0.12832392  0.02769325   4.6338 4.07e-06 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Coefficients	Expected sign	Actual sign
lvr	+	+
ref	-	-
insur	-	-
rate	+	+
amount	+	+
credit	-	-
term	+/-	-
arm	+/-	+

These signs are all reasonable.

b.

Call:

```
glm(formula = delinquent ~ lvr + ref + insur + rate + amount +
     credit + term + arm, family = binomial(link = "logit"), data = lasvegas)
```

Coefficients:

```
      Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.665303   1.992157   0.836 0.403195
lvr           0.013519   0.008199   1.649 0.099164 .
ref          -0.525064   0.230491  -2.278 0.022725 *
insur        -3.122105   0.216906 -14.394 < 2e-16 ***
rate          0.308656   0.082771   3.729 0.000192 ***
amount        0.217606   0.113769   1.913 0.055786 .
credit       -0.003564   0.001933  -1.844 0.065232 .
term         -0.133037   0.034597  -3.845 0.000120 ***
arm           1.405649   0.371355   3.785 0.000154 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 998.03 on 999 degrees of freedom
Residual deviance: 666.29 on 991 degrees of freedom
AIC: 684.29
```

Number of Fisher Scoring iterations: 5

The signs of the estimated coefficients are the same as the linear probability model, except of the significance. (Logit model: lvr, amount and credit are insignificant)

c.

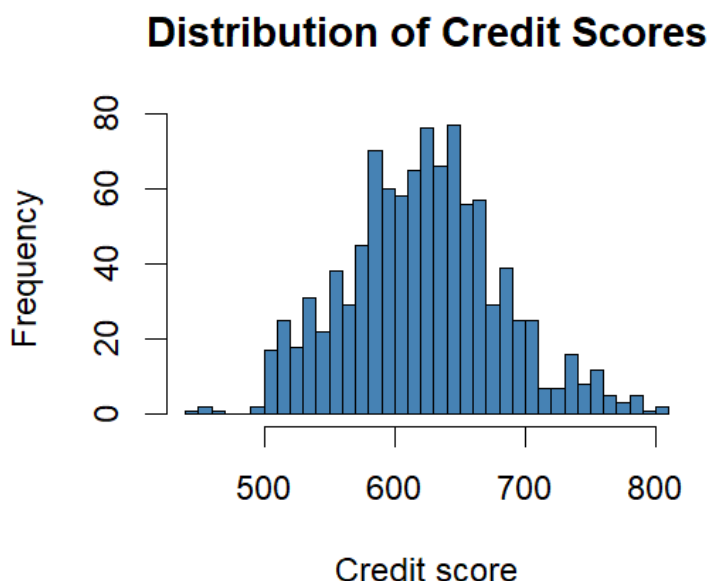
```
> # Linear probability model predictions
> pred_lpm <- predict(lpm, type = "response")
> pred_lpm[c(500, 1000)]
      500      1000
0.1827828 0.5785296
> # Logit predicted probabilities
> pred_logit <- predict(logit, type = "response")
> pred_logit[c(500, 1000)]
      500      1000
0.1335848 0.6265535
```

The linear probability model predicts that the 500th obs has 18.2% estimated probability to be DELINQUENT. The 1000th obs has 57.8% estimated probability to be DELINQUENT.

The logit predict model predicts that the 500th obs has 13.4% estimated probability to be DELINQUENT. The 1000th obs has 62.7% estimated probability to be DELINQUENT.

d.

```
> # LPM predictions
> pred_d_lpm <- predict(lpm, newdata = newdata)
> names(pred_d_lpm) <- c("credit=500", "credit=600", "credit=700")
> pred_d_lpm
credit=500 credit=600 credit=700
0.5532956 0.5091060 0.4649165
> # Logit predictions
> pred_d_logit <- predict(logit, newdata = newdata, type = "response")
> names(pred_d_logit) <- c("credit=500", "credit=600", "credit=700")
> pred_d_logit
credit=500 credit=600 credit=700
0.4967030 0.4086308 0.3260580
```



Similarity: as the credit score increases, both models predict that the probability of delinquency increases.

Difference: The predicted probability of the linear probability model is higher than the logit model in terms of credit score (500, 600, 700)

e.

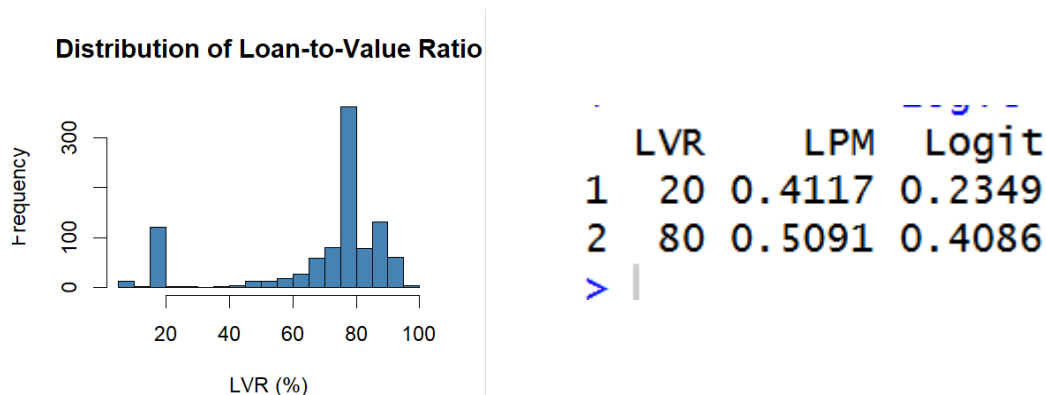
```
> # LPM: marginal effect is just the coefficient (constant)
> lpm_credit_coef <- coef(lpm)["credit"]
> cat("LPM marginal effect of CREDIT:", lpm_credit_coef, "\n")
LPM marginal effect of CREDIT: -0.0004418953
```

The ME of linear probability model is -0.000442, implying that credit score increases 100 would increase the probability of delinquency 4.42%

```
> me_logit
      credit=500      credit=600      credit=700
-0.0008910620 -0.0008613439 -0.0007832567
```

The ME of logit model decrease from -0.000891 to -0.000783, implying that credit score increases 100 when credit score =500 would increase the probability of delinquency 8.91%; and that credit score increases 100 when credit score =700 would increase the probability of delinquency 7.83%

f.



LVR=20% and LVR=80%, the estimated probability of the probability models are both higher than the estimated probability of the logit model.

g.

```
> # LPM
> pred_lpm_class <- ifelse(pred_lpm >= 0.5, 1, 0)
> lpm_correct <- mean(pred_lpm_class == lasvegas$delinquent)
> cat("LPM correct predictions:", round(lpm_correct * 100, 2), "%\n")
LPM correct predictions: 85.8 %
> # Logit
> pred_logit_class <- ifelse(pred_logit >= 0.5, 1, 0)
> logit_correct <- mean(pred_logit_class == lasvegas$delinquent)
> cat("Logit correct predictions:", round(logit_correct * 100, 2), "%\n")
Logit correct predictions: 85.4 %
> # Confusion matrices
> table(Predicted = pred_lpm_class, Actual = lasvegas$delinquent)
      Actual
Predicted 0    1
0      727  68
1      74 131
> table(Predicted = pred_logit_class, Actual = lasvegas$delinquent)
      Actual
Predicted 0    1
0      736  81
1      65 118
```

h.

```
> t(results)
      threshold accuracy false_neg_rate
[1,]      0.10    0.614      0.1641791
[2,]      0.15    0.752      0.2686567
[3,]      0.20    0.808      0.3134328
[4,]      0.25    0.830      0.3134328
[5,]      0.30    0.854      0.3432836
[6,]      0.35    0.862      0.3731343
[7,]      0.40    0.868      0.3880597
[8,]      0.45    0.878      0.4179104
[9,]      0.50    0.880      0.4925373
[10,]     0.55    0.872      0.5970149
[11,]     0.60    0.874      0.6567164
[12,]     0.65    0.880      0.7014925
[13,]     0.70    0.872      0.8208955
[14,]     0.75    0.862      0.9104478
[15,]     0.80    0.870      0.9552239
[16,]     0.85    0.870      0.9552239
[17,]     0.90    0.868      0.9850746
>
```

By testing different thresholds, we can see that threshold has the highest accuracy 0.88.

Threshold = 0.65 also gives accuracy 0.88, however its false negative rate is 0.7, which is higher than threshold=0.5 values. Therefore threshold=0.5 is still a better rule.